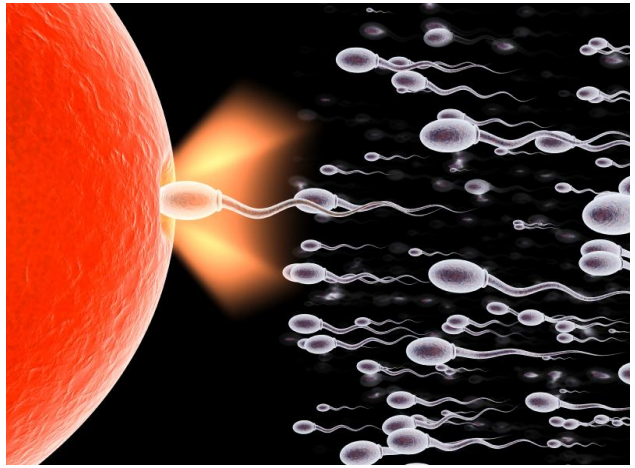
FERTILIZATION & FIRST WEEK OF DEVELOPMENT





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Science & Technology للعلوم والتكنولوجيا

SLOs

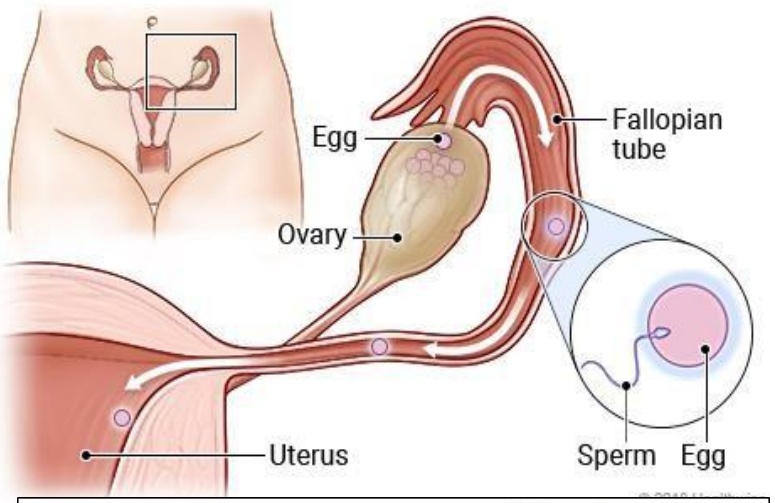
- **SLO# 1:** Define fertilization and prerequisites for fertilization
- **SLO# 2:** Describe the phases of fertilization
- **SLO# 3:** Explain the Effects / Consequences of fertilization
- **SLO# 4:** Describe polyspermy and how it can be prevented
- **SLO# 5:** Describe the formation of zygote, morula and blastocyst

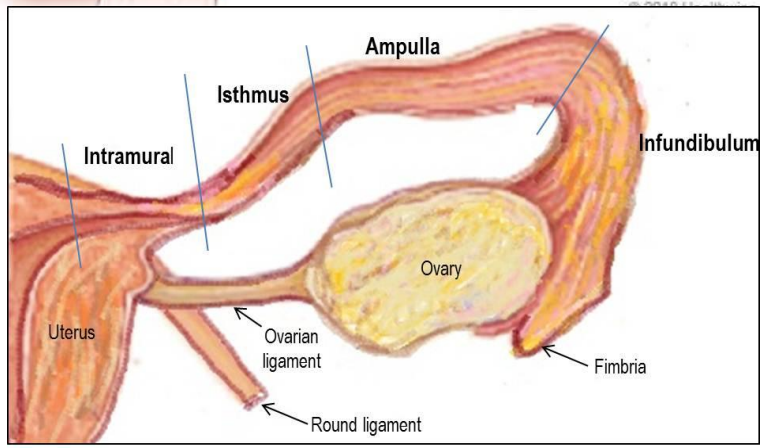
FERTILIZATION

Definition: It is the process in which the male gamete – **SPERMATOZOA** fuses with the female gamete **- OVA** to form the **ZYGOTE.**

- **Time**: around midcycle (**day 14**) time of ovulation
- **Site**: usually, **ampulla** of **uterine tube** (the widest part).
- **Prerequisites:** Some events must occur before fertilization to allow for it, They include:

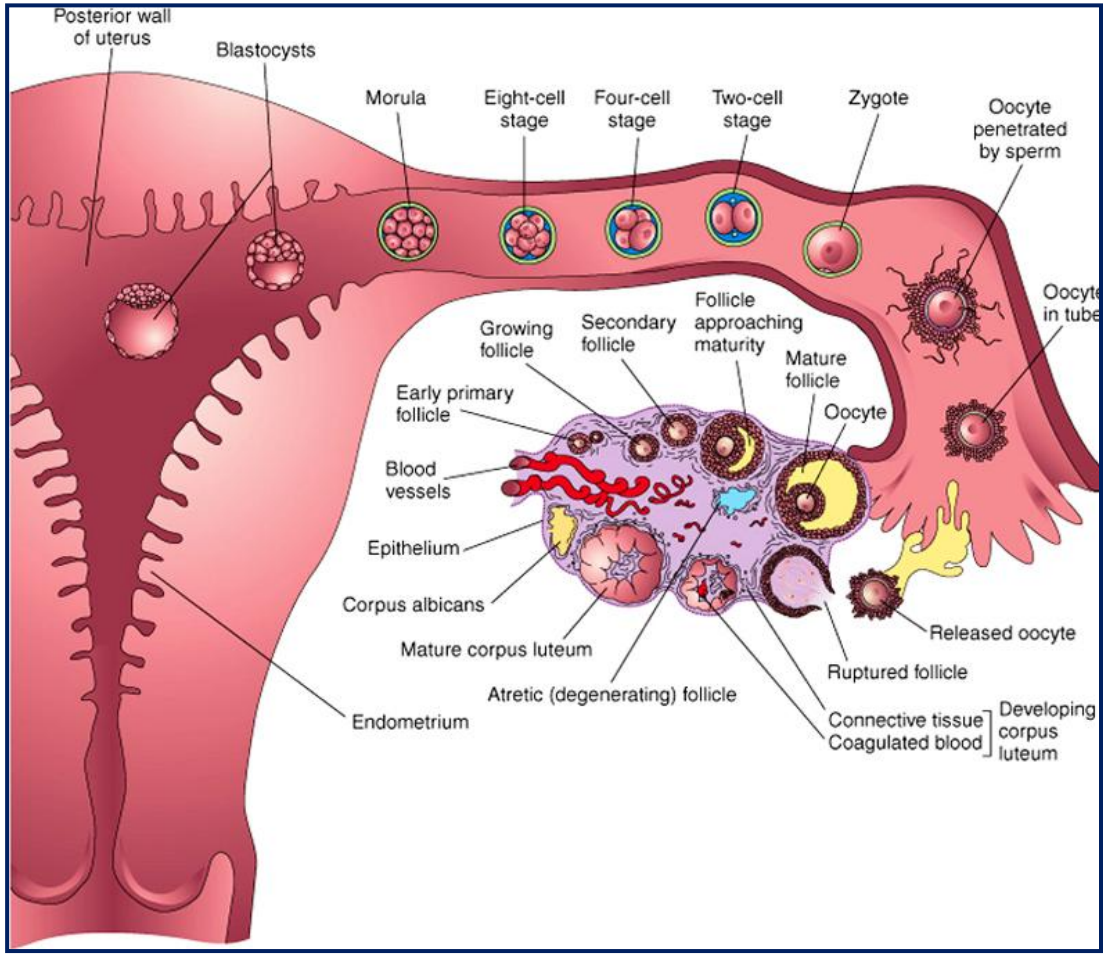
1. **Sperm and oocyte transport**
2. **Capacitation**
3. **Acrosome reaction**





1a. OOCYTE TRANSPORT

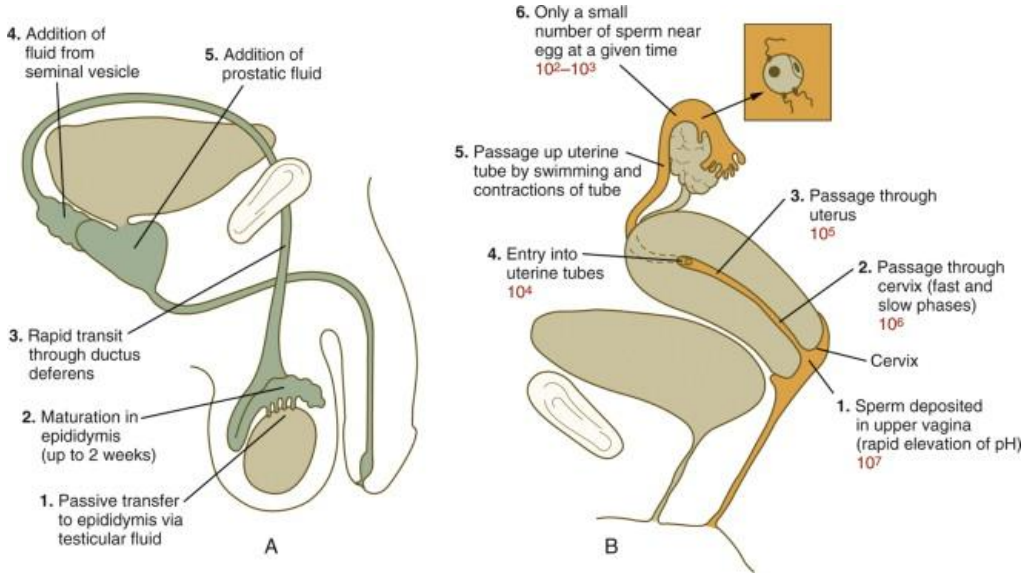
- During ovulation, secondary oocyte is **expelled** from ovary to **uterine tube**.
- Fluid currents produced by **cilia** and **peristalsis** of the tube; sweep the oocyte towards fertilization site (ampulla)



1b. SPERM TRANSPORT

In male:

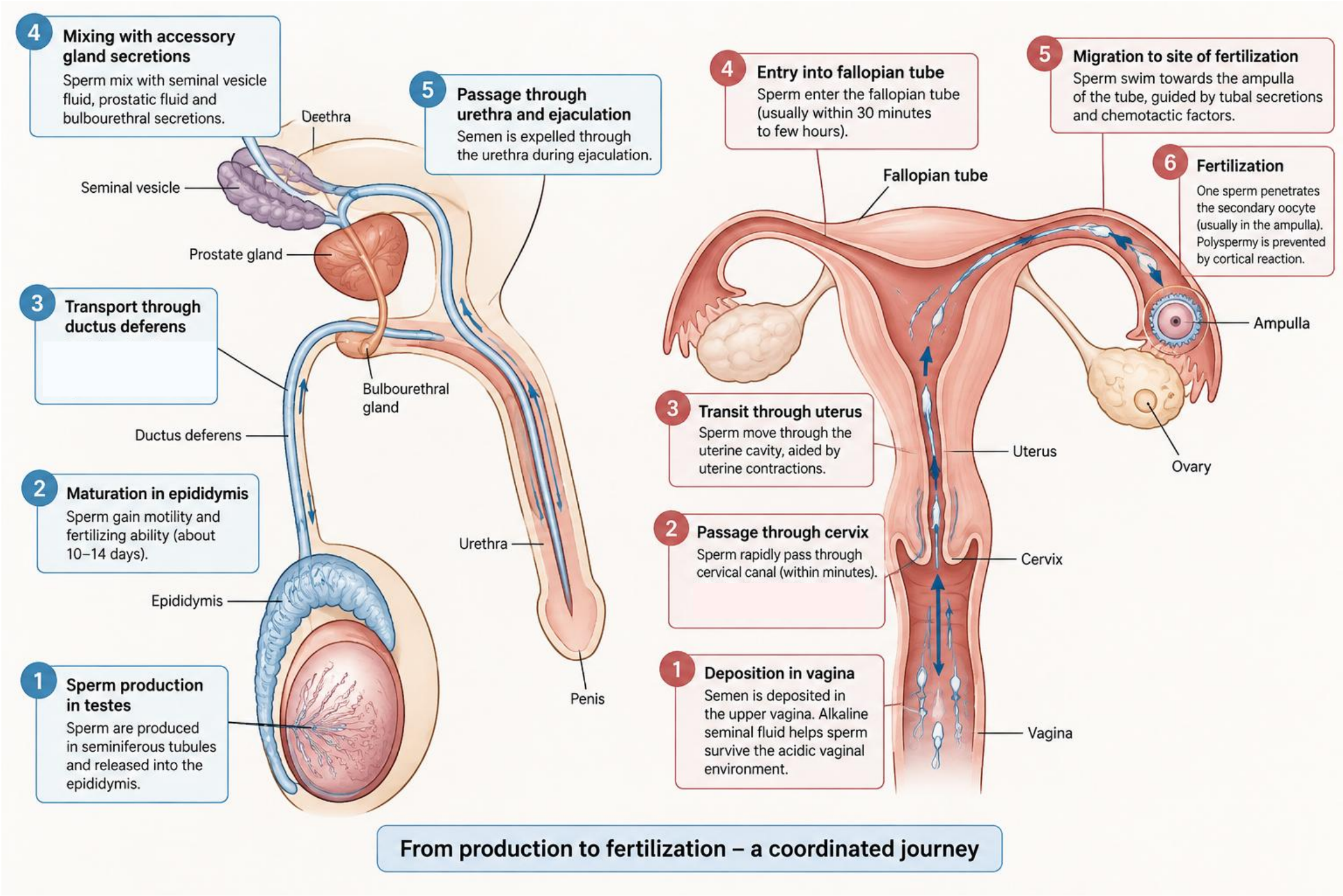
- from storage site mainly in tail of epididymis → by **peristaltic contractions** of vas deferens → urethra.



In female:

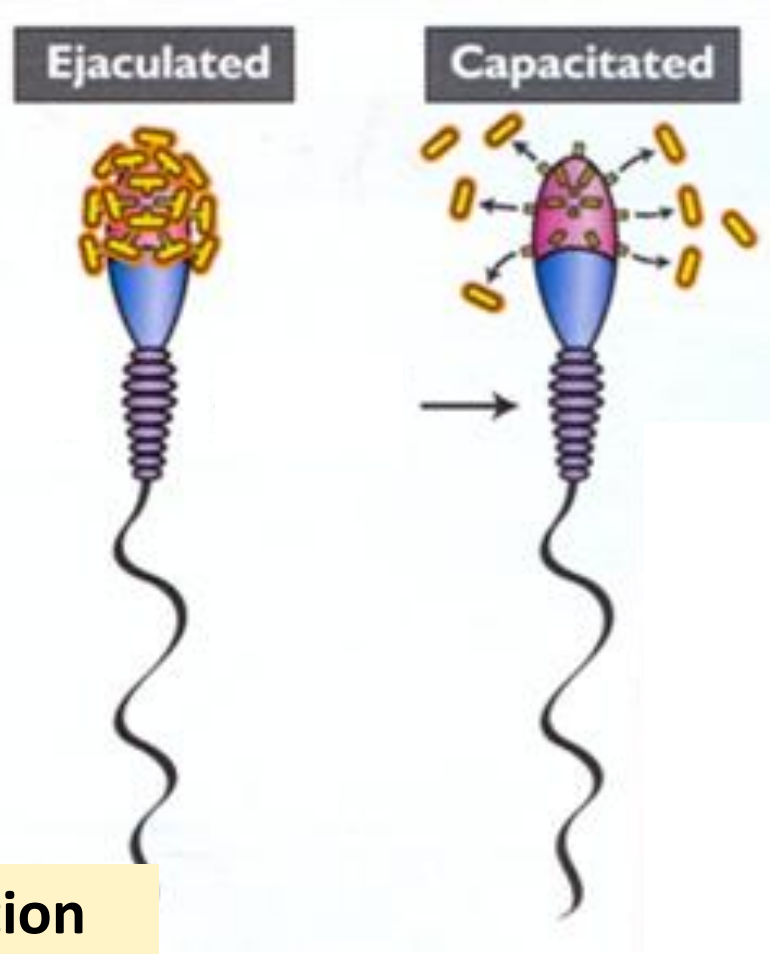
- During sexual intercourse, 200-600 million sperms are ejaculated and **deposited** in posterior fornix of vagina. Many sperms die due to vaginal acidity
- Sperms pass through cervix by movements of their tails but slowly due to cervical mucus
- Passage of sperms through uterus & uterine tubes by their muscular contractions
- **Fate:** 200 sperms reach ampulla within 5 minutes
- The fertilization power of sperms last for 2-5 days.

NICE TO KNOW


4 Mixing with accessory gland secretions
Sperm mix with seminal vesicle fluid, prostatic fluid and bulbourethral secretions.
5 Passage through urethra and ejaculation
Semen is expelled through the urethra during ejaculation.
Deethra
Seminal vesicle
Prostate gland
3 Transport through ductus deferens
Bulbourethral gland
Ductus deferens
2 Maturation in epididymis
Sperm gain motility and fertilizing ability (about 10–14 days).
Urethra
Epididymis
1 Sperm production in testes
Sperm are produced in seminiferous tubules and released into the epididymis.
Penis
4 Entry into fallopian tube
Sperm enter the fallopian tube (usually within 30 minutes to few hours).
5 Migration to site of fertilization
Sperm swim towards the ampulla of the tube, guided by tubal secretions and chemotactic factors.
Fallopian tube
6 Fertilization
One sperm penetrates the secondary oocyte (usually in the ampulla). Polyspermy is prevented by cortical reaction.
Ampulla
3 Transit through uterus
Sperm move through the uterine cavity, aided by uterine contractions.
Uterus
Ovary
2 Passage through cervix
Sperm rapidly pass through cervical canal (within minutes).
Cervix
1 Deposition in vagina
Semen is deposited in the upper vagina. Alkaline seminal fluid helps sperm survive the acidic vaginal environment.
Vagina
From production to fertilization – a coordinated journey

2. CAPACITATION

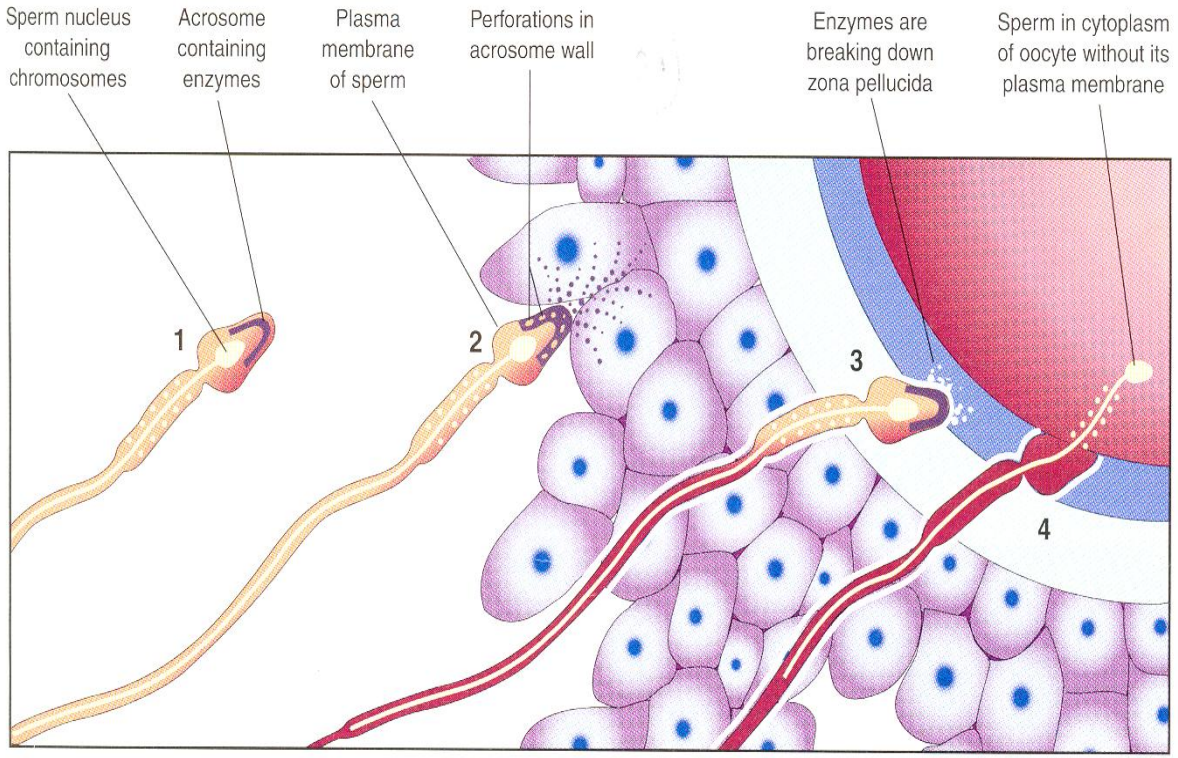
- **Definition:** Removal of proteins from the plasma membrane overlying the head of sperm
- **Results**: The sperm becomes more active and motile and ready to fertilize ovum.
- **Duration**: 5- 7 hours.
- **Site**: starting at cervix of uterus (continues through female genital tract).



Note: The capacitated sperms are only capable of fertilization

3. ACROSOMAL REACTION

- **Definition:** secretion of enzymes by acrosome to help penetration of egg coats
- **Site:** surrounding oocyte
- **Enzymes released:**
  - a) Hyaluronidase: penetrates corona radiate.
  - b) Trypsin-like substances: digest zona pellucida.
  - c) Zona lysin/acrosin: soften zona pellucida.



NB. Only capacitated sperm can undergo acrosome reaction

PHASES OF FERTILIZATION

1) Penetration of Corona Radiata:

A. The enzyme **hyaluronidase** produced by **acrosome**.

B. Tubal mucosal enzymes

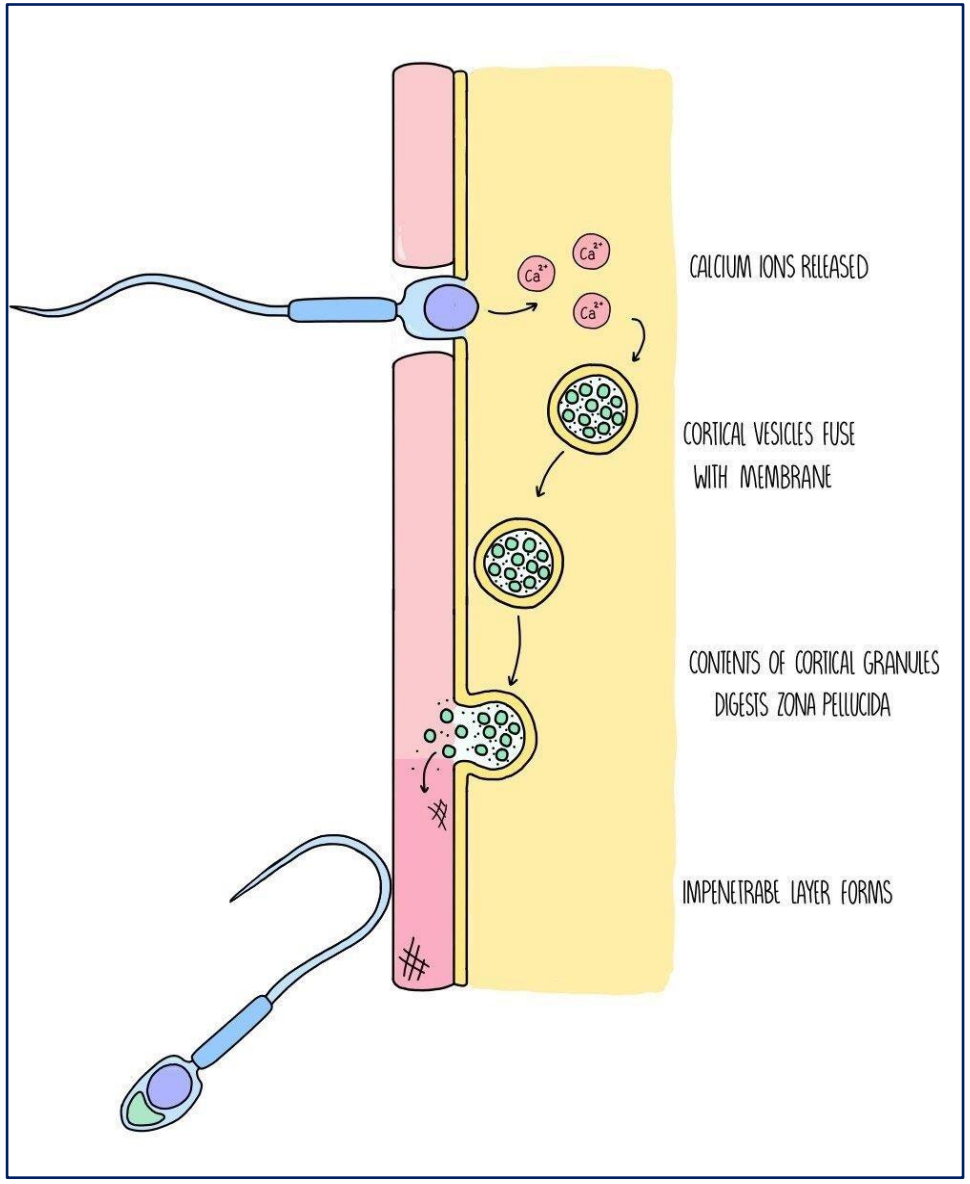
C. Movements of sperm tails.

All three assist the spermatozoa to pass through – to reach zona pellucida

2) Penetration of Zona pellucida:

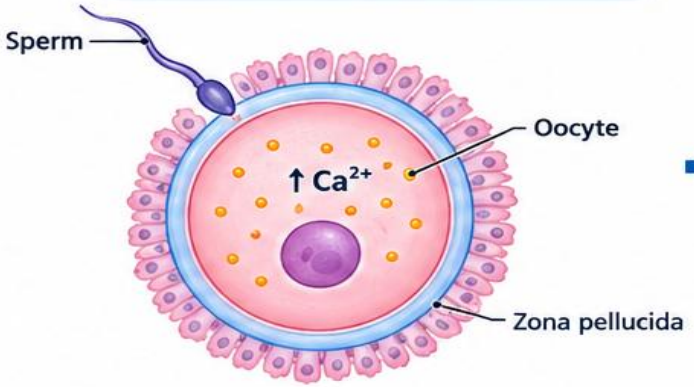
1. Acrosome Reaction
2. Zona Reaction

- **Once one sperm has successfully entered, the zona alters its composition.**
- **It becomes impermeable to another sperm.**

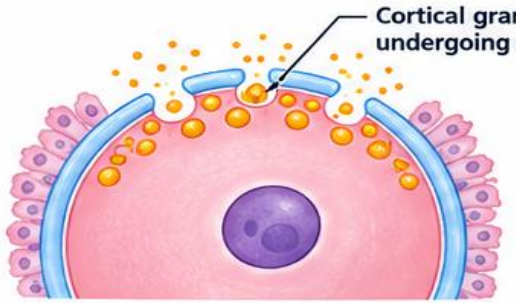


Cortical Reaction & Zona Reaction

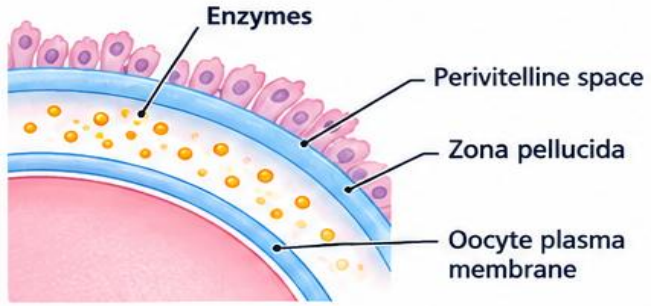
1 Sperm–oocyte fusion triggers Ca^{2+} rise in the oocyte.



2 Cortical granules beneath the oocyte plasma membrane undergo exocytosis.

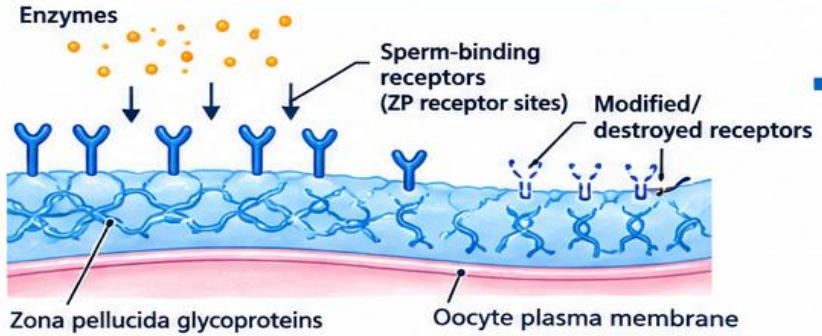


3 Cortical granules release hydrolytic/modifying enzymes into the perivitelline space.

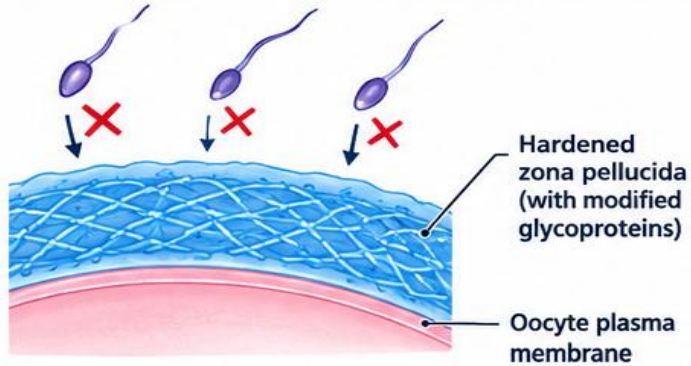


Cortical reaction

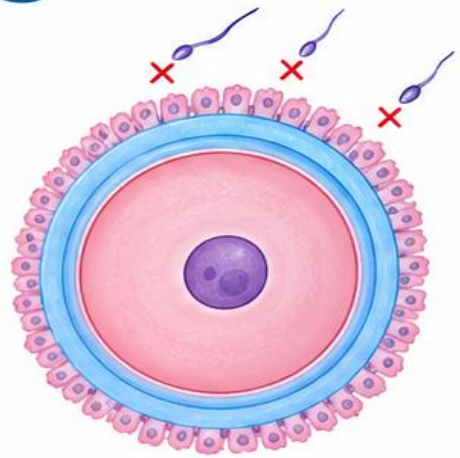
4 Enzymes modify zona pellucida glycoproteins and sperm-binding receptors (including ZP receptor sites), causing receptor destruction/modification.



5 Zona pellucida becomes hardened and less permeable to additional sperm.



6 Result: block to polyspermy.



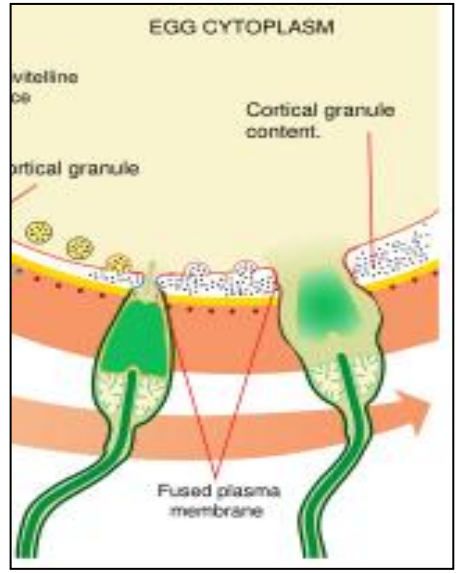
Zona reaction

Prevents polyspermy

PHASES OF FERTILIZATION

3) Fusion of Plasma membrane of Sperm & Oocyte:

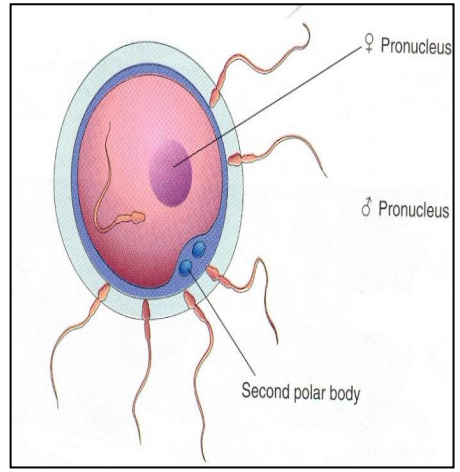
- The plasma membrane of sperm & oocyte fuse.
- The membranes **breakdown** at the **site of fusion**.
- The **sperm head** & **tail enters the oocyte** leaving behind the plasma membrane.



4) Activation of oocyte (completion of Meiosis II of the Oocyte):

After sperm entry, **Secondary oocyte completes 2nd** meiotic division ➔mature ova & 2nd polar body.

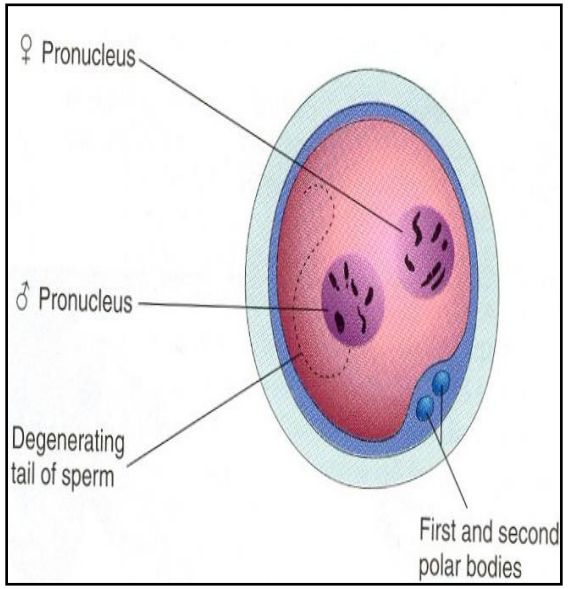
The mature ova enlarge by decondensation of chromosomes & form **FEMALE PRONUCLEUS.(22,X)**



PHASES OF FERTILIZATION

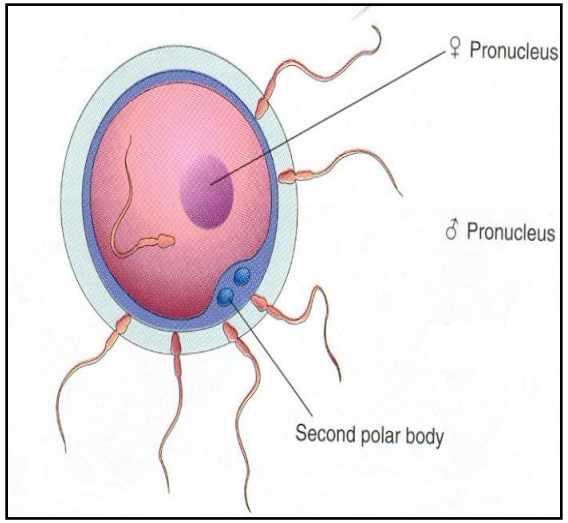
5) Formation of Male pro nucleus:

- The nucleus of the sperm increases within the oocyte cytoplasm.
- The tail degenerates.
- **MALE PRONUCLEUS** is formed.(22,X or Y)
- The oocyte containing two haploid pronuclei is called as **OOTID.**

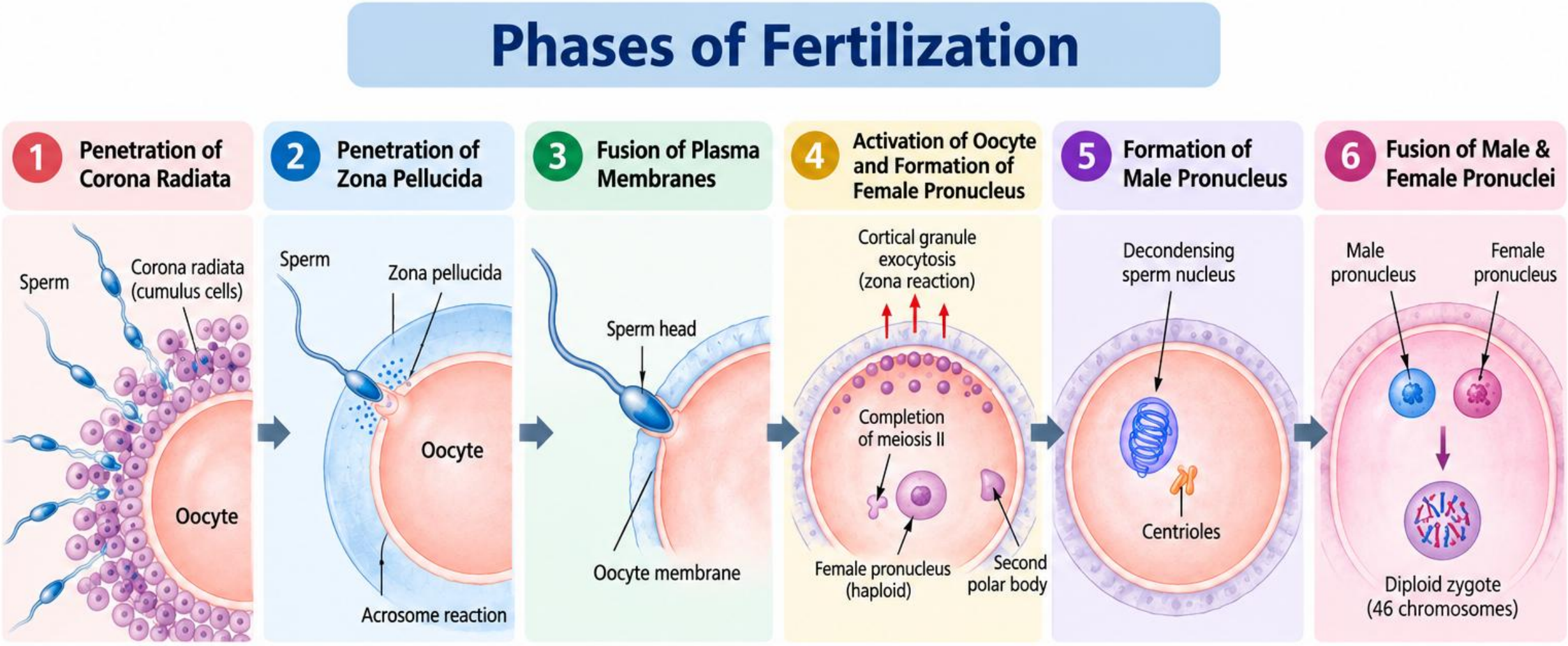


6) Fusion of Male & Female pro nucleus :

- The two pronuclei fuse into a single diploid aggregation of chromosomes.
- The ootid is now called as **ZYGOTE**.
- Both the pronuclei before fusion replicate their DNA so that two chromatids are attached to each centromere.



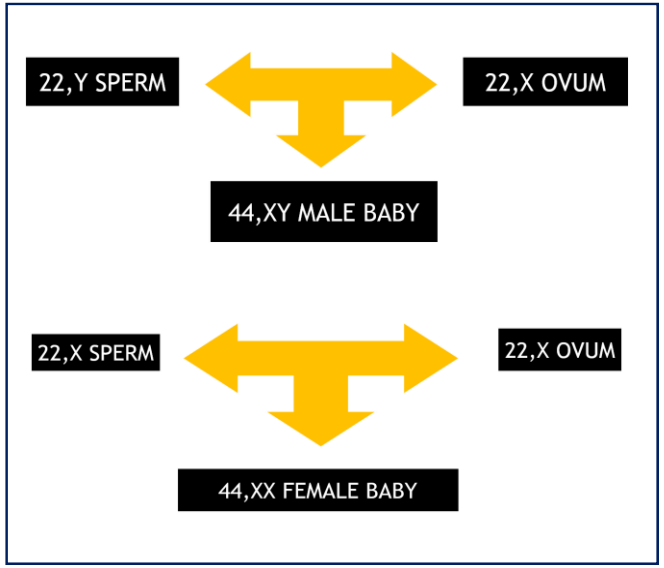
SUMMARY


Phases of Fertilization
1
Penetration of Corona Radiata
Sperm
Corona radiata (cumulus cells)
Oocyte
2
Penetration of Zona Pellucida
Sperm
Zona pellucida
Oocyte
Acrosome reaction
3
Fusion of Plasma Membranes
Sperm head
Oocyte membrane
4
Activation of Oocyte and Formation of Female Pronucleus
Cortical granule exocytosis (zona reaction)
Completion of meiosis II
Female pronucleus (haploid)
Second polar body
5
Formation of Male Pronucleus
Decondensing sperm nucleus
Centrioles
6
Fusion of Male & Female Pronuclei
Male pronucleus
Female pronucleus
Diploid zygote (46 chromosomes)

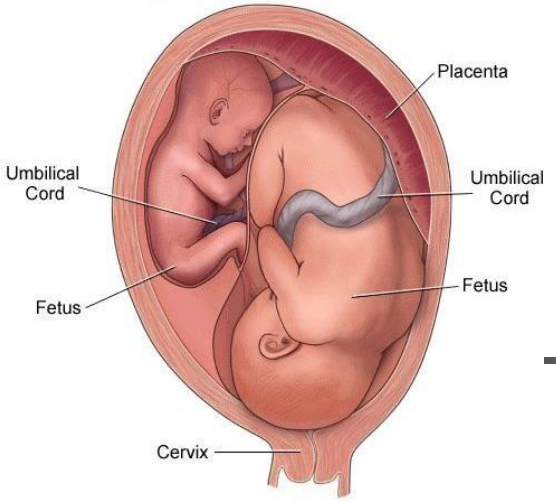
RESULTS (EFFECTS/ CONSEQUENCE) OF FERTILIZATION

1. Stimulates Secondary oocyte to complete 2nd meiotic division.
2. Cortical and zona reaction
3. Restores normal diploid number of chromosomes (46) in zygote.
4. Determines sex of embryo; X sperm produces a female & Y sperm produces a male.
5. Results in variation of human species by thorough shuffling of maternal & paternal chromosomes.
6. Initiation of cell division of zygote (cleavage)

SEX OF THE BABY



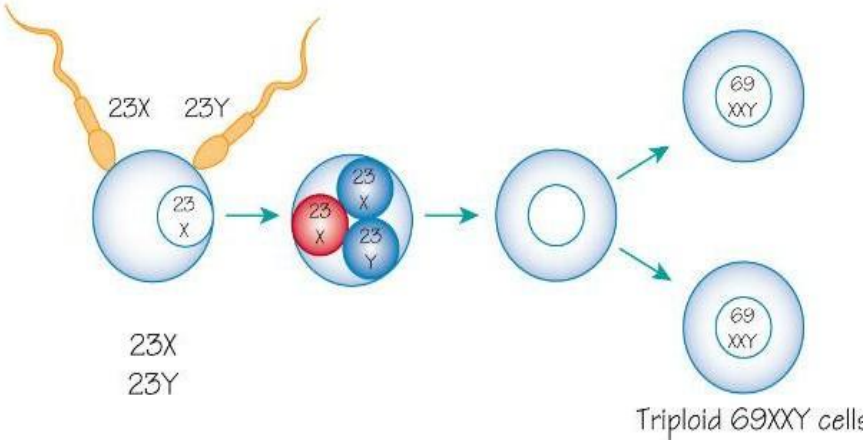
ABNORMAL FERTILIZATION



1) Superfecundation:

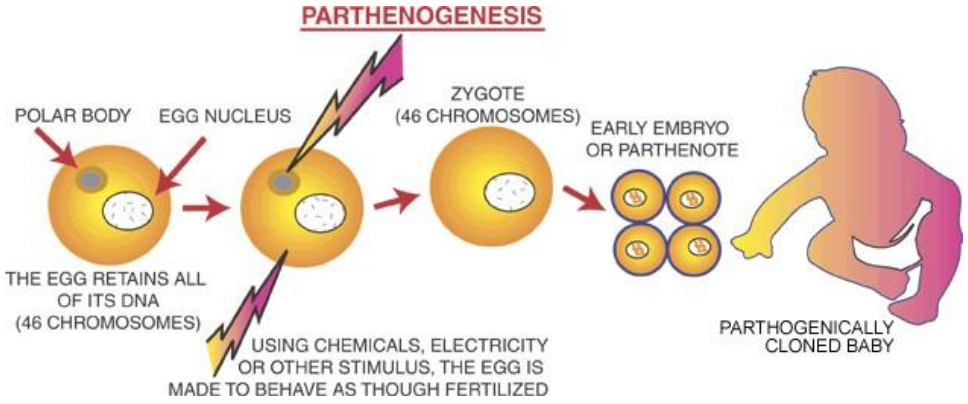
*Fertilization of 2 or more ova released during same menstrual cycle by sperms from separate acts of coitus from same or another male

NB: If the ova are from different cycles➔ **Superfetation**



2) Polyspermia

*Fertilization by 2 sperms, results in a zygote with extra set of chromosomes (69 chrmosomes = 3n).



3) Parthenogesis (virgin birth):

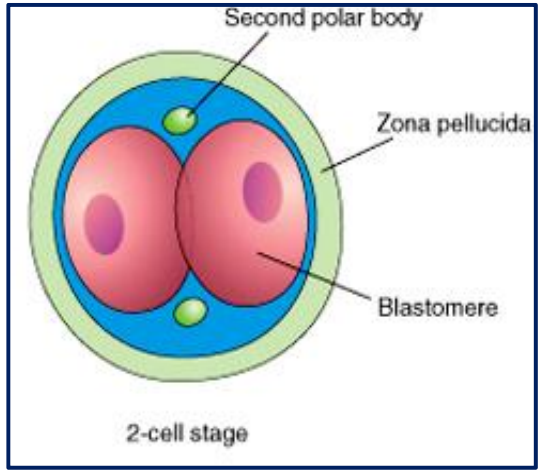
*Cleavage of an unfertilized oocyte to give a growing embryo without being fertilized by a male

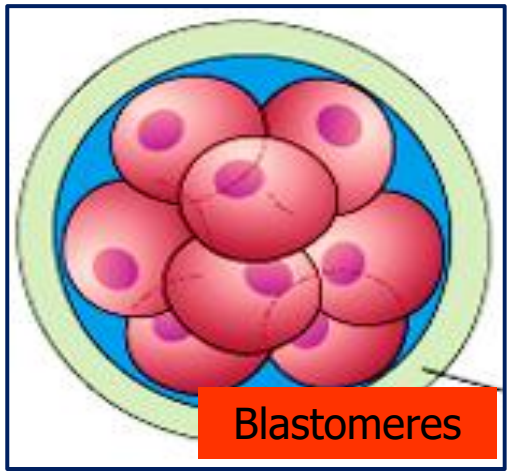
*It occurs naturally in plants, invertebrates & fish

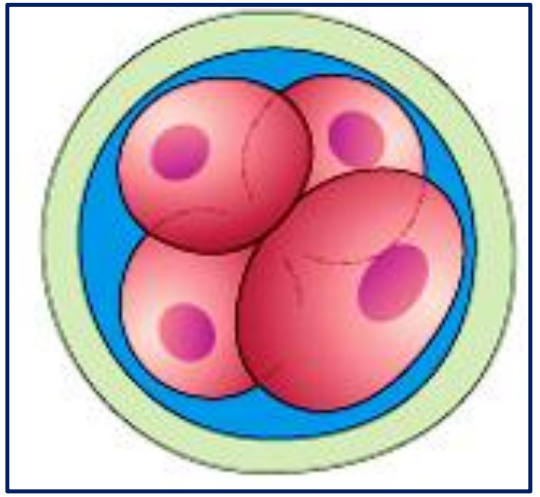
CLEAVAGE

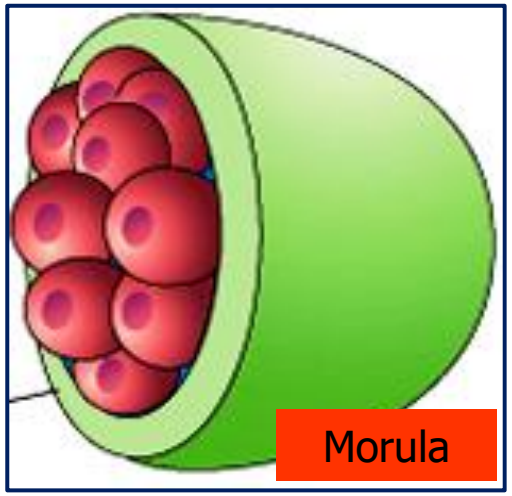
- After fertilization the zygote immediately starts its mitotic divisions.
- The onset of mitosis in the zygote is called cleavage.
- It commences **within 30 hrs** after fertilization.

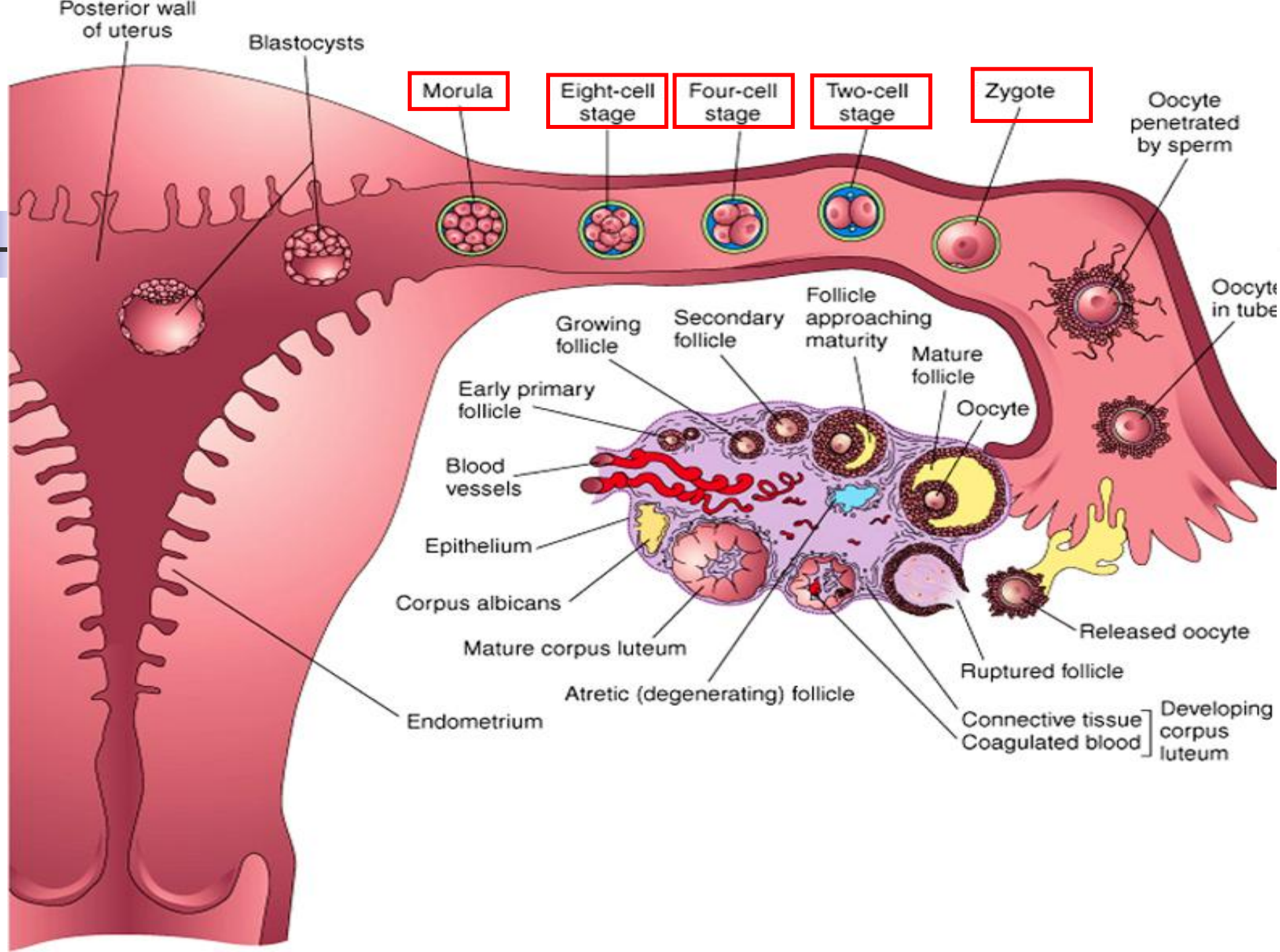
- **Site: in uterine tube.**
- **Duration: 1st 3 days**
- **The divided cells (blastomeres) (2 → 4 → 8 → 16 cells) with smaller cytoplasm.**
- **Zona pellucida is still surrounding blastomeres**





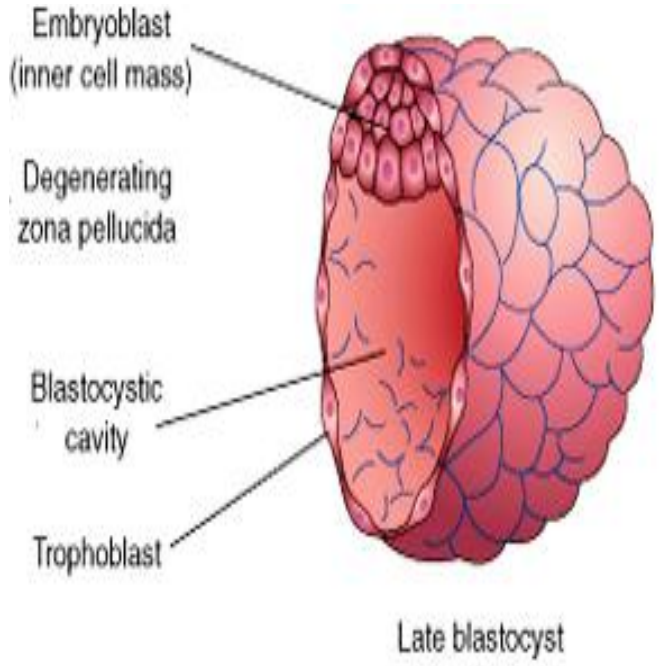





Posterior wall of uterus
Blastocysts
Morula
Eight-cell stage
Four-cell stage
Two-cell stage
Zygote
Oocyte penetrated by sperm
Oocyte in tube
Growing follicle
Secondary follicle
Follicle approaching maturity
Mature follicle
Early primary follicle
Oocyte
Blood vessels
Epithelium
Corpus albicans
Mature corpus luteum
Atretic (degenerating) follicle
Endometrium
Released oocyte
Ruptured follicle
Connective tissue
Coagulated blood
Developing corpus luteum

BLASTOCYST

- Shortly after the **morula** reaches the uterus, fluid passes from the uterine cavity through the zona pellucida into the morula.
- Soon fluid filled spaces appears inside the morula.
- These coalesce & form a cyst – **BLASTOCYST.**
- The fluid filled cavity located inside the blastocyst is called as **BLASTOCOEL.**

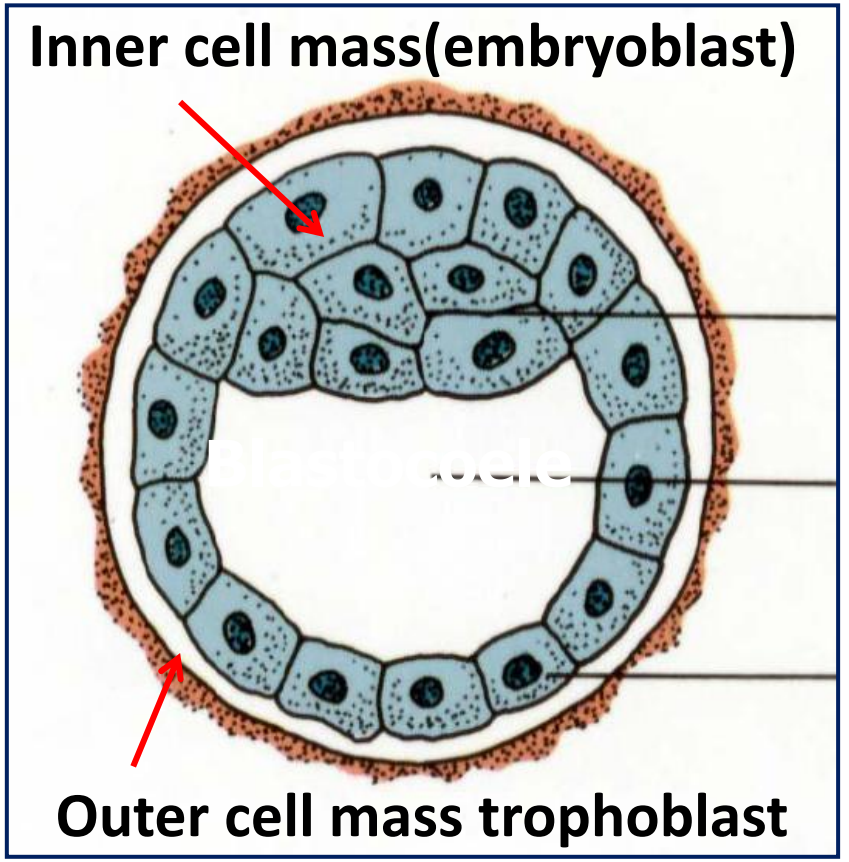


Late blastocyst

Note: The blastocyst is the entire multi-cellular structure in early mammalian development, while the blastocoel is the fluid-filled cavity located within the blastocyst.

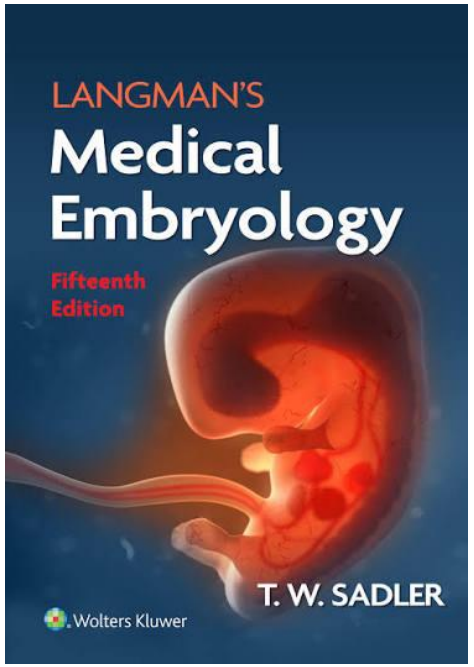
Embryo pole

- As more & more fluid enters the blastocyst it pushes the inner cell mass to an eccentric position.
- The outer layer of flattened cells lining the blastocele – TROPHOBLAST.
- It forms the fetal part of placenta.
- The inner cell mass forms the primordium of the embryo – EMBRYOBLAST.
- Blastocyst Now is ready for implantation



Abembryonic pole

REFERENCES



Langman's Medical Embryology 15th Edition:

Chapter: Fertilization and First week of development

"The important thing is not to stop questioning.

Curiosity has its own reason for existing."

